Exact Kernels, Specialization Defects, and Odd-Degree Rigidity in an Explicit Affine-Cancellation Construction

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M107

October 8, 2026

Abstract

The source manuscript claims a complex affine fourfold whose cylinder is affine five-space although the fourfold is not affine space. We separate its explicit coordinate identities from its all-derivation degeneration, lifting, and rigidity argument. For a nonconstant polynomial ϕ, we compute the complete kernel of the associated ambient locally nilpotent derivation by an exact special-fiber saturation certificate. Kernel formation fails to commute with reduction: the resulting special-fiber inclusion has fraction-field degree $\deg \phi$, and even in degree one misses a boundary divisor geometrically. We give finite orbit-analysis orders, extend the homogeneous rigidity obstruction to every odd-degree ϕ of degree at least three and all fiber shifts below two, and prove a higher-root version of the extraction lemma. These are algebraic local results, not numerical certificates for the source's global cancellation claim. Classical slice, saturation, and polynomial *abc* tools are credited; confidential finite-moment originals are used only within their finite scope. Contribution assessment: C3/5, a notable source-specific kernel and specialization calculation with modular rigidity extensions, not an independent resolution of complex affine-space cancellation.

Contents

1 Scope, status, and the source dependency map **2**
1.1 What is imported and what is proved here 2

2 A polynomial family and its ambient derivation **3**
2.1 The source's stable-coordinate identity is retained 4

3 The complete ambient kernel **4**

4 An exact specialization defect **5**

5 Finite orbit analysis and its Ram comparison **6**

6 Odd-degree homogeneous rigidity **7**
6.1 Sharpness and the even-degree boundary 8

7 Extraction for higher-root modifications **9**

8 Source checks and the global boundary 10

9 Advantages, limitations, and further work 11

10 Evidence and contribution assessment 11

1 Scope, status, and the source dependency map

All rings below have characteristic zero. A locally nilpotent derivation (LND) D on a domain B satisfies: for each $f \in B$ there is an integer $n(f)$ with $D^{n(f)}f = 0$. This is a pointwise, not a uniform, nilpotence condition. A coordinate in $B = \mathbb{C}^{[n]}$ is an element belonging to a full polynomial coordinate system. A one-stable coordinate becomes a coordinate after adjoining one independent variable. Neither an LND kernel on an open set nor a finite coefficient-rank test decides these global properties.

The assigned cached manuscript [1], dated September 23, 2026, asserts $A[w] \simeq \mathbb{C}^{[5]}$ and $A \not\simeq \mathbb{C}^{[4]}$ for a specific four-dimensional domain A. We call these *source claims*, not established consequences of our finite tests. All ten cached TeX files and the source README were literally read: 2,299 lines in eleven files. This includes the complete degeneration, bundle-lifting, reduction, rigidity, and fibration arguments, not just an introduction or a slice lemma. Reading a proof is not independent verification of every cited geometric input or of worldwide acceptance of its conclusion.

The current primary-literature check has a narrower role. Gaifullin and Petrov's July 2026 introduction [2] distinguishes the generalized cancellation problem from affine-space cancellation and describes the characteristic-zero affine-space problem as open at that time. That predates the assigned September manuscript; it neither refutes nor validates it. Dutta–Lahiri, Theorem 3.4 [3], states that a residual coordinate in $R^{[n]}$, $n \geq 3$, is one-stable when R is a one-dimensional affine algebra over an algebraically closed field and either the field has characteristic zero or R_{red} is seminormal. Its exact statement and proof were inspected. It explains why a residual coordinate can be stably tractable without deciding that it is already a coordinate. We use no claim that the new cancellation assertion has received independent community verification.

1.1 What is imported and what is proved here

The seven scientific source modules have the following roles.

1. `01-introduction`: states cancellation, stable-coordinate, and nontrivial affine-fibration claims; fixes their distinction.
2. `02-construction`: gives the hypersurface, localized coordinates, an ambient LND, and an explicit cylinder coordinate change. We retain these formulas and prove the family identities used below.
3. `03-degeneration`: identifies the entire associated graded ring, not merely a convenient subring, and induces a nonzero highest derivation. Its all-generator reduction is a necessary global input.
4. `04-bundle`: principalizes the center on a $\mathbb{G}_m$-torsor and lifts arbitrary additive actions via line-bundle linearization. Lifting cannot be replaced by the one ambient LND studied here.
5. `05-reduction`: extracts a homogeneous LND on the torsor ring from a positive-degree invariant. Section 7 proves a generalized algebraic extraction statement, conditional on that setup.

6. `06-rigidity`: uses a second grading, polynomial *abc*, and an original-field nonsquare obstruction. Section 6 proves an odd-degree and lower-term extension of this local argument.

7. `07-affine-fibrations`: proves fiber and smoothness statements and uses locally-polynomial/symmetric-algebra results and projective-module freeness. These geometric imports are not proved by our kernel computation or by our exact tests.

The independently proved local increments are Theorems 3.2, 4.1, and 6.2, and Proposition 7.1. No new global noncancellation family is asserted for general ϕ or general root exponent. In particular the companion M109 work is not used as a premise.

2 A polynomial family and its ambient derivation

Let $\phi \in \mathbb{C}[u]$ be nonconstant of degree $m \geq 1$, and put $P = \mathbb{C}[p, s, u, F, J]$. Define

$$x = s^2 + \phi(u) + p^2 F, \qquad y = s + x(x - \phi(u)), \qquad z = sx + p^2 J, \tag{1}$$

$$H_\phi = x^2 F - (1 + 2sx)J - p^2 J^2 - pu, \qquad h_\phi = H_\phi + pu. \tag{2}$$

The source is the specialization $\phi(u) = u^3$. The symbol h_ϕ here is an invariant in the ambient ring; it is not the source's cylinder variable or its quotient coordinate u.

Lemma 2.1. *There are exact identities*

$$xy - z(z+1) = p^2 h_\phi, \qquad P_p = \mathbb{C}[p, p^{-1}, x, y, z, u]. \tag{3}$$

The derivation $\Delta_\phi = -p^2 \partial_u$ in the coordinates on the right preserves P, is locally nilpotent there, and satisfies

$$\begin{aligned} &\Delta p = 0, && \Delta u = -p^2, && \Delta s = -p^2 x \phi'(u), \\ &\Delta F = (2sx+1)\phi'(u), && \Delta J = x^2 \phi'(u), && \\ &\Delta x = \Delta y = \Delta z = \Delta h_\phi = 0, && \Delta H_\phi = p^3. && \end{aligned} \tag{4}$$

Proof. Expand $xy = sx + x^2(s^2 + p^2 F)$ and $z(z+1) = s^2 x^2 + sx + p^2(2sx+1)J + p^4 J^2$. Their difference is exactly $p^2 h_\phi$. The inverse coordinates are

$$s = y - x(x - \phi(u)), \quad F = p^{-2}(x - s^2 - \phi(u)), \quad J = p^{-2}(z - sx). \tag{5}$$

Differentiating these expressions with p, x, y, z fixed gives (4). All five images belong to P. Every element of P_p is a polynomial in u over $\mathbb{C}[p^{\pm 1}, x, y, z]$, so sufficiently many applications of $-p^2 \partial_u$ kill it. Restricting to P gives an LND. The last identity follows from $H_\phi = h_\phi - pu$. □

A crucial non-descent. Δ_ϕ does *not* induce a derivation on $A_\phi = P/(H_\phi)$: $\Delta_\phi(H_\phi) = p^3$ is not in (H_ϕ). Indeed the localization of the quotient is a polynomial ring over $\mathbb{C}[p^{\pm 1}]$, so p^3 remains nonzero there. Its exponential in an extra variable instead sends H_ϕ to $H_\phi + p^3 w$. This is the source's mechanism for comparing cylinders, not an action on A_ϕ.

2.1 The source's stable-coordinate identity is retained

For $\phi = u^3$, set $x_0 = s^2 + u^3$ and

$$L = x_0^2 F - (1 + 2sx_0)J, \qquad M = (1 - 2sx_0)F + 4s^2 J.$$

This is a determinant-one change in (F, J). Thus $P = \mathbb{C}[p, s, u, M][L]$, and the source writes

$$H = L - pu + p^2 Q(L) + p^4 F(L)^3, \quad Q = 2x_0 F^2 - 2sFJ - J^2.$$

With $C = Q(0)$ and $L_* = pu - p^2 C$, $H(L_*)$ is divisible by p^3. On the cylinder equation $H + p^3 w = 0$ the source constructs a polynomial e with $L = L_* + p^3 e$; substitution gives $w = -H(L_* + p^3 e)/p^3$ polynomially. The exact divisibility and cancellation in a domain matter: a formula valid only after inverting p would not give a global cylinder isomorphism. This report credits that construction to the source and does not claim it as a new finite-moment result.

3 The complete ambient kernel

The local slice identifies the kernel away from $p = 0$ immediately. The additional point here is a complete, rather than degree-truncated, saturation certificate at the exceptional fiber.

Lemma 3.1 (One-element saturation)**.** *Let $B \subseteq K \subseteq P$ be domains, let $p \in B$ be nonzero, and suppose $B_p = K_p$ and $B \cap pP = pB$. Then $B = K$.*

Proof. For $f \in K$ there is $n \geq 0$ with $p^n f \in B$. If $n > 0$, that element lies in $B \cap pP = pB$, so $p^n f = pb$ with $b \in B$. Cancellation gives $p^{n-1} f = b$. Induction removes all denominators. The intersection condition is exactly injectivity of $B/pB \longrightarrow P/pP$. □

This classical lemma and the local-slice mechanism also occur in the freshly inspected private comparison [6]; the new content is their explicit complete application to (1).

Theorem 3.2 (Global kernel for every nonconstant ϕ)**.** *The assignment $(X, Y, Z, t) \mapsto (x, y, z, h_\phi)$ induces an isomorphism*

$$\ker_P \Delta_\phi \simeq B := \mathbb{C}[p, X, Y, Z, t]/(p^2 t - XY + Z(Z + 1)). \tag{6}$$

For every $n \geq 1$, its embedded image satisfies $B \cap p^n P = p^n B$. In particular the answer is not only a presentation after localization or in a finite-degree vector space.

Proof. The displayed defining polynomial is primitive and linear in t over the UFD $\mathbb{C}[p, X, Y, Z]$, since p does not divide $XY - Z(Z + 1)$. It is irreducible by Gauss's lemma, so B is a domain. After inverting p its map to P_p identifies it with $\mathbb{C}[p^{\pm 1}, x, y, z] = \ker_{P_p} \Delta_\phi$. As B has no p-torsion, the map is injective before localization.

It remains to prove the intersection condition. Write

$$\xi = s^2 + \phi(u), \quad L = \xi^2 F - (1 + 2s\xi)J, \quad M = (1 - 2s\xi)F + 4s^2 J.$$

The coefficient matrix from (F, J) to (L, M) has determinant $4s^2\xi^2 + (1 + 2s\xi)(1 - 2s\xi) = 1$. Hence

$$P/pP = \mathbb{C}[s, u, L, M], \quad F = 4s^2 L + (1 + 2s\xi)M, \quad J = -(1 - 2s\xi)L + \xi^2 M. \tag{7}$$

The reduction of B is the domain $B_0 = \mathbb{C}[X, Y, Z, t]/(XY - Z(Z + 1))$, mapping by

$$X \mapsto \xi, \quad Z \mapsto s\xi, \quad Y \mapsto s + s^2\xi, \quad t \mapsto L. \tag{8}$$

On inverting X, use $S = Z/X$ to write $B_{0,X} = \mathbb{C}[X^{\pm 1}, S, t]$. Its map sends (X, S, t) to (ξ, s, L). These are algebraically independent: a polynomial in X over $\mathbb{C}[s, L]$ of positive degree d has, after substituting $s^2 + \phi(u)$, a nonzero leading term of u-degree md. The constant case is also injective. Since B_0 is a domain and $X \neq 0$, injectivity after localization implies injectivity before it. Lemma 3.1 now proves (6). Repeated application of $B \cap pP = pB$ proves the assertion for every p^n as well. □

What completeness costs. No search up to a guessed degree bound appears in this proof. The essential global step is exact injectivity on the special fiber. A finite collection of invariant generators with the correct generic transcendence degree would not, by itself, establish that step. Nor does (6) compute the intersection of kernels of *all* LNDs on the quotient A_ϕ.

4 An exact specialization defect

The preceding proof gives an injective reduction of the global kernel. It does not say that every invariant of the reduced action lifts.

Theorem 4.1 (Degree and boundary of the reduced inclusion)**.** *For $m = \deg \phi \geq 1$, in the coordinates (7),*

$$\overline{\Delta}_\phi = \phi'(u)\partial_M, \qquad \ker \overline{\Delta}_\phi = \mathbb{C}[s, u, L]. \tag{9}$$

The natural injection

$$(\ker \Delta_\phi)/p(\ker \Delta_\phi) = B_0 \hookrightarrow \ker \overline{\Delta}_\phi$$

has fraction-field degree m. Over $X \neq 0$ the corresponding map is finite free of rank m, with the extension obtained by adjoining a root of $\phi(U) - (X - S^2)$. It is not an isomorphism globally for any m: the image on spectra misses the nonempty divisor $X = 0, Z = -1$ of Spec B_0.

Proof. Modulo p, the derivation fixes s, u, ξ. Formula (4) gives

$$\overline{\Delta} L = 0, \qquad \overline{\Delta} M = ((1 - 2s\xi)(1 + 2s\xi) + 4s^2\xi^2)\phi'(u) = \phi'(u).$$

Since ϕ' is a nonzero polynomial and the ring is a characteristic-zero domain, the kernel of $\phi'(u)\partial_M$ is exactly the polynomial subring independent of M. The fraction fields in question are

$$\mathbb{C}(s, \xi, L) \subseteq \mathbb{C}(s, u, L), \qquad \xi = s^2 + \phi(u).$$

Over $K = \mathbb{C}(s, L)$, the polynomial $\phi(U) - Y$ is irreducible in $K[U, Y]$: it is linear in Y and its quotient ring is $K[U]$. It is primitive as a polynomial over $K[Y]$, so it remains irreducible over $K(Y)$ by Gauss's lemma. Its U-degree is m, proving the degree claim after replacing Y by $\xi - s^2$.

More concretely, on $X \neq 0$ the target algebra is

$$\mathbb{C}[X^{\pm 1}, S, t][U]/(\phi(U) - (X - S^2)).$$

Dividing by the nonzero leading coefficient makes the relation monic; $1, U, \ldots, U^{m-1}$ is a free basis. Identifying $S = s, t = L$ gives $\mathbb{C}[s, u, L, (s^2 + \phi(u))^{-1}]$, as required. Finally, at any point of the target with $X = \xi = 0$, (8) forces $Z = s\xi = 0$. But $X = 0, Z = -1$ satisfies $XY = Z(Z + 1)$ with arbitrary Y, t; it is a nonempty divisor and has no preimage. An isomorphism could not miss it. □

Example 4.2 (The cubic and the linear edge case)**.** For the source $\phi = u^3$, the invariant u of the reduced derivation does not even belong to Frac B_0: it has degree three over it. The closed-fiber kernel is nevertheless a polynomial ring in three variables. For $\phi = u$, the field degree is one, but u is not in B_0. Indeed in the birational coordinates one would have $u = X - (Z/X)^2$, which has a pole along $X = 0, Z = -1$. Thus fraction-field equality is weaker than equality of the coordinate rings, even in the simplest member of the family.

A useful computational separation. On $X \neq 0$, the derivative $\phi'(U)$ detects the possible ramification of this finite free algebra. It is also precisely the coefficient of the reduced derivation. This is an exact relation between the polynomial cover and the speed of the reduced additive action, not a numerical rank heuristic. The failure to cover the other boundary component is a second, distinct obstruction; it persists even when ϕ' is a unit.

5 Finite orbit analysis and its Ram comparison

The original confidential finite-moment material [5] was freshly read at lines 155–180 and 455–476. Its finite analysis map $m = Ec$ is invertible on its range when the chosen matrix has full column rank; its positive weighted Gram construction is confined to a fixed finite polynomial space. It supplies no theorem that finite rank implies completeness of a global invariant algebra. The separately read slice comparison [6], lines 290–364 and 398–424, is explicitly distinguished from that original material.

Here the orbit parameter supplies a natural finite coefficient bank, with an exact stopping order dictated by the source coordinates. Let $q = -u/p^2 \in P_p$, so $\Delta q = 1$, and define

$$\Theta(f) = \sum_{j \geq 0} \frac{(-q)^j}{j!} \Delta^j f. \tag{10}$$

The sum is finite for each f. It is evaluation at $u = 0$ while fixing p, x, y, z, hence is an algebra homomorphism into $\ker_{P_p} \Delta$. In particular

$$\Theta(s) = y - x(x - \phi(0)), \quad \Theta(F) = p^{-2}(x - \Theta(s)^2 - \phi(0)), \quad \Theta(J) = p^{-2}(z - x\Theta(s)).$$

Taylor synthesis is the exact finite identity

$$f = \sum_{j \geq 0} \frac{q^j}{j!} \Theta(\Delta^j f). \tag{11}$$

For a monomial in q this is ordinary Taylor reconstruction at zero; linearity proves it for all of $P_p = (\ker \Delta)_p[q]$.

Proposition 5.1 (An exact orbit-order budget)**.** *For nonnegative integers a, b, c, d, e and $m = \deg \phi \geq 1$,*

$$\operatorname{ord}_\Delta(p^a s^b u^c F^d J^e) = mb + c + 2md + me, \tag{12}$$

where $\operatorname{ord}_\Delta f = \max\{j : \Delta^j f \neq 0\}$ for $f \neq 0$. Consequently a finite span of such monomials with the right side at most D requires at most $D + 1$ orbit coefficients in (11). For linear combinations the order may decrease by cancellation, but never increases beyond this maximum.

Proof. In the localized coordinates, s has u-degree m with leading coefficient $x \operatorname{lc}(\phi)$. Formula (5) gives F degree $2m$ with leading coefficient $-p^{-2}x^2 \operatorname{lc}(\phi)^2$, and J degree m with leading coefficient $-p^{-2}x^2 \operatorname{lc}(\phi)$. None is zero. Degrees add for products in this domain, and $-p^2\partial_u$ has nilpotence order equal to the polynomial's u-degree in characteristic zero. □

The bound is a usable finite stopping certificate, not a new abstract Taylor theorem. Its denominator p^{-2j} is real and cannot be ignored near the exceptional fiber. Numerical conditioning of a moment matrix would depend on its basis, scaling, and norm; no such matrix or condition number is assumed here. Equations (10)–(11) can verify invariance in a specified finite bank. Theorem 3.2 requires the separate all-order saturation proof and is not inferred from that bank.

6 Odd-degree homogeneous rigidity

To avoid confusing the variable c with a leading coefficient, write the latter as $\gamma \in \mathbb{C}^*$. Put

$$R = \mathbb{C}[a, d, b, c, u]/(ac - bd - 1), \qquad \mathrm{wt}(a, d, b, c, u) = (1, 1, -1, -1, 0). \tag{13}$$

This is a domain, for example by localization at a and the primitive linear defining equation. Its fraction field is $\mathbb{C}(a, d, b)(u)$, with u transcendental over $\mathbb{C}(a, d, b)$. An LND E has fiber shift $e \in \mathbb{Z}$ if it sends the weight-k piece to the weight-$(k + e)$ piece for every k.

We first record the polynomial *abc* estimate in precisely the form used.

Lemma 6.1 (Polynomial *abc*)**.** *Let $A + B = C$ be nonzero pairwise coprime polynomials over a characteristic-zero field, not all constant. If r is the number of distinct roots of ABC in an algebraic closure, then* $\max(\deg A, \deg B, \deg C) \le r - 1$.

Proof. The Wronskian $A'B - AB'$ is nonzero: otherwise A/B is constant and coprimality makes both constant, and then so is C. At a root of any one of A, B, C of multiplicity k, this Wronskian is divisible by the $(k-1)$st power of its linear factor. For C use $A'C - AC' = A'B - AB'$. Coprimality keeps the three sets of roots disjoint. Thus its degree is at least $\deg A + \deg B + \deg C - r$, and at most $\deg A + \deg B - 1$. This bounds $\deg C$ by $r - 1$; permuting the three polynomials bounds the others. Passing to an algebraic closure is only for counting roots, not for changing the field of coefficients in the later nonsquare argument. □

Theorem 6.2 (Odd-degree and lower-auxiliary-term obstruction)**.** *Fix an odd integer $m \ge 3$ and $\gamma \in \mathbb{C}^*$. Give R the auxiliary grading*

$$\deg_{\mathrm{aux}}(a, d, b, c, u) = (0, m, -m, 0, 2), \qquad N = d^2 + \gamma a^2 u^m. \tag{14}$$

Let $W \in R$ have fiber weight two and every auxiliary component of W have degree strictly less than $2m$. Set $v = -N + W$. There is no nonzero fiber-homogeneous LND E of shift $e < 2$ such that $E^2(v) = 0$. In particular this applies to

$$v_\phi = a^3 b - a^2 \phi(u) - d^2 \tag{15}$$

for every polynomial ϕ of odd degree $m \ge 3$ and leading coefficient γ.

Proof. The relation $ac - bd - 1$ is auxiliary homogeneous of degree zero, so (14) is an actual grading, not just a weight on representatives. It commutes with fiber weight. A derivation on this finitely generated graded algebra has finitely many homogeneous auxiliary components. Let E' be the highest nonzero component of E. It has the same fiber shift e and is an LND: if $(E')^k f$ is nonzero for homogeneous f, it is the highest possible component of $E^k f$; eventual vanishing of the latter forces eventual vanishing of the former. The highest part of v is $-N$, so $E^2 v = 0$ implies $(E')^2 N = 0$.

Use the orbit parameter q and polynomials $a(q) = \exp(qE')a$, $d(q) = \exp(qE')d$, $u(q) = \exp(qE')u$ in $K[q]$, where $K = \mathrm{Frac}\, R$. Then

$$d(q)^2 + \gamma a(q)^2 u(q)^m = N(q), \qquad \deg_q N(q) \le 1. \tag{16}$$

Each factor and $N(q)$ is nonzero since its value at $q = 0$ is nonzero. The two summands are coprime: a common root would be a root of d and of a or u, giving multiplicity at least two in both summands, hence in the nonzero polynomial $N(q)$ of degree at most one. It follows that the three polynomials in (16) are pairwise coprime.

Let D_a, D_d, D_u denote their factor degrees. If either summand is nonconstant, its degree is at least two. Cancellation above degree one therefore forces both summands to have equal degree

$$M = 2D_d = 2D_a + mD_u \geq 2.$$

Their number of distinct roots, together with those of $N(q)$, is at most $D_d + D_a + D_u + 1$. Lemma 6.1 gives

$$2D_d \leq D_d + D_a + D_u, \qquad (m-2)D_u \leq 0. \tag{17}$$

Thus $D_u = 0$ and $D_d = D_a > 0$. The leading coefficients δ, α of $d(q), a(q)$ must satisfy $\delta^2 + \gamma\alpha^2 u^m = 0$, since $u(q) = u$. This would make $-\gamma u^m$ a square in the *original* field $K = \mathbb{C}(a, d, b)(u)$. Its u-adic valuation is the odd integer m, whereas a square has even valuation, a contradiction. Consequently both summands are constant, which forces $D_a = D_d = D_u = 0$. Therefore E' fixes a, d, u.

Differentiating $ac - bd = 1$ now gives $aE'c = dE'b$. The element $h = cE'b - bE'c \in R$ satisfies

$$E'b = ah, \qquad E'c = dh. \tag{18}$$

Localize the fixed subring $C_0 = \mathbb{C}[a, d, u]$ at all its nonzero elements. The result is $Q[b]$, where $Q = \operatorname{Frac} C_0$, because $c = (1 + bd)/a$. A nonzero Q-derivation $f(b)\partial_b$ of $Q[b]$ is locally nilpotent only when f is constant. Indeed for degree at least two the degrees of successive iterates of b strictly grow with nonzero leading coefficients, and for degree one its nonzero slope persists in every iterate. Hence (18) implies $h \in Q$ (also if $h = 0$).

In fact $R \cap Q = C_0$. The a-chart is $R_a = \mathbb{C}[a^{\pm 1}, d, u, b]$, so the intersection there is $C_0[a^{-1}]$; the d-chart is $R_d = \mathbb{C}[a, d^{\pm 1}, u, c]$, giving $C_0[d^{-1}]$. The intersection of these two coefficient rings in Q is C_0, since the independent primes a, d cannot occur in a common reduced denominator. Thus $h \in C_0$. But h is fiber homogeneous of weight $e - 2 < 0$, and C_0 has only nonnegative weights. Therefore $h = 0$, making E' zero on all five generators, a contradiction. Finally, for (15), $W = a^3 b - a^2(\phi(u) - \gamma u^m)$ has precisely the required lower degrees. □

6.1 Sharpness and the even-degree boundary

The source itself records the LND

$$E_+(a) = E_+(d) = E_+(u) = 0, \qquad E_+(b) = a, \quad E_+(c) = d. \tag{19}$$

It preserves $ac - bd - 1$, has fiber shift two, and is triangular over $\mathbb{C}[a, d, u]$. For every ϕ, $E_+ v_\phi = a^4 \neq 0$ but $E_+^2 v_\phi = 0$. Thus the bound $e < 2$ in Theorem 6.2 is exact for the family (15). This example is retained from the source, not credited as a new construction; the extension is the larger forbidden range and odd-degree lower-term class. For arbitrary W, (19) need not square-kill W, so no universal sharpness assertion is made for that larger class.

For even $m = 2k \geq 2$, there is an explicit failure of the intermediate claim that $(E')^2 N = 0$ forces E' to fix a, d, u. Take $\gamma = 1$, $\kappa = iu^k$, and define

$$E_0 u = 0, \qquad E_0 a = \kappa a + d, \qquad E_0 d = -\kappa^2 a - \kappa d,$$
$$E_0 b = \kappa b + c, \qquad E_0 c = -\kappa^2 b - \kappa c.$$

The same trace-zero square-zero matrix acts on (a, d) and (b, c). It preserves their determinant, fixes its coefficients, and its exponential is polynomial; hence it gives an LND on R of fiber shift zero. Set $w = d + \kappa a$. Then

$$E_0 w = 0, \quad N = (d + \kappa a)(d - \kappa a), \quad E_0 N = -2\kappa w^2 \neq 0, \quad E_0^2 N = 0.$$

Nevertheless this is *not* an LND satisfying $E_0^2 v_{u^m} = 0$. At the valid point $a = 1, d = b = 0, c = 1, u = 1$ one obtains $E_0^2(a^3 b) = 6i \neq 0$; the other summand has second derivative zero. This distinguishes failure of the leading-form obstruction from a counterexample to the full fiber polynomial's rigidity.

7 Extraction for higher-root modifications

We isolate one further modular consequence of the source's reduction. It explains which part of the argument actually needs the quadratic root exponent and which does not. It constructs no geometric torsor.

Let R now be any characteristic-zero domain with an integer fiber grading, and let $0 \neq v \in R$ be fiber homogeneous of weight $d > 0$. For an integer $r \geq 2$ put

$$G_r = R[t, V]/(t^r V - v), \quad \deg(R, t, V) = (0, -1, r), \quad \mathrm{wt}(t) = 0, \quad \mathrm{wt}(V) = d. \tag{20}$$

The degree and fiber gradings are distinct. There is an injection $G_r \hookrightarrow R[t, t^{-1}]$ sending V to v/t^r. For completeness, if $tF = (t^r V - v)Q$ in $R[t, V]$, reducing modulo t gives $v\overline{Q} = 0$ and hence $Q = tQ_1$. Cancellation gives $F = (t^r V - v)Q_1$. Thus the quotient has no t-torsion and injects into its localization, a domain.

Proposition 7.1 (A uniform square-killing extraction)**.** *Let $D \neq 0$ be an LND on G_r, homogeneous of degree shift j and fiber shift zero. Suppose* $\ker D$ *contains a nonzero homogeneous element of positive degree. Then $DV = 0$ and there is a nonzero fiber-homogeneous LND E on R with fiber shift at most zero and $E^2 v = 0$. If $Dt \neq 0$, necessarily*

$$j - 1 = r\ell, \quad \ell \geq 0, \qquad D(b) = t^{r-1} V^{\ell+1} E(b) \quad (b \in R), \tag{21}$$

and E has strictly negative fiber shift $-d(\ell + 1)$. If $Dt = 0$, one can choose E with $Ev = 0$.

Proof. For any nonzero f in a characteristic-zero domain with LND D, set $\nu(f) = \max\{n : D^n f \neq 0\}$. Leibniz's formula and its unique top nonzero term give $\nu(fg) = \nu(f) + \nu(g)$ and $\nu(Df) = \nu(f) - 1$ when $Df \neq 0$. In particular invariant factors of a nonzero invariant product are invariant. Also $Df \notin fG_r$ when $Df \neq 0$, since otherwise its order would be at least $\nu(f)$.

The homogeneous pieces, obtained by reducing every factor $t^r V$ to v, are

$$(G_r)_n = \begin{cases} t^{-n} R, & n < 0, \\ t^h V^k R, & n \geq 0, \quad k = \lceil n/r \rceil, \quad h = rk - n \in \{0, \ldots, r-1\}. \end{cases} \tag{22}$$

For $n = 0$ this means $k = h = 0$. Uniqueness follows in the Laurent domain. Every positive-degree homogeneous element is divisible by V. The assumed invariant and the order identity therefore imply $DV = 0$.

First suppose $Dt \neq 0$. The element Dt has degree $j - 1$ and cannot belong to tG_r. Formula (22) shows that all pieces other than nonnegative multiples of r are divisible by t. Hence $j - 1 = r\ell$ with $\ell \geq 0$, and the degree-j formula gives (21). The common multiplier is nonzero, so the product rule shows that E is a derivation of R. Fiber degrees give its stated shift. It is nonzero: if D killed R and V, differentiating $t^r V = v$ would force $Dt = 0$.

Let $m_0 = \nu(t) \geq 1$. For $E(b) \neq 0$, (21) gives

$$\nu(Eb) = \nu(b) - \big((r-1)m_0 + 1\big). \tag{23}$$

Successive applications strictly decrease a nonnegative integer until they vanish; thus E is locally nilpotent on R. Moreover $\nu(v) = rm_0$ and $Dv = rt^{r-1} V Dt \neq 0$. Hence $\nu(Ev) = m_0 - 1$. If $E^2 v \neq 0$, its order would be

$$(m_0 - 1) - \big((r-1)m_0 + 1\big) = (2 - r)m_0 - 2 < 0,$$

impossible. Therefore $E^2 v = 0$ for every $r \geq 2$, not merely for $r = 2$.

If $Dt = 0$, both t, V are invariant. The degree-j piece in (22) has the form $c_0 R$ with $c_0 = t^h V^k$, $k \geq 0$ (for negative j use $k = 0, h = -j$). Write $D(b) = c_0 E(b)$. This gives a nonzero derivation E on R, and $D^n b = c_0^n E^n b$ because $Dc_0 = 0$; consequently E is an LND. Differentiating the relation gives $Dv = 0$, whence $Ev = 0$. Its fiber shift is $-dk \leq 0$. □

Corollary 7.2. *For R as in (13), v as in Theorem 6.2, and any $r \geq 2$, the ring G_r has no nonzero LND that is homogeneous for the degree grading, has fiber shift zero, and has a nonzero positive-degree homogeneous invariant.*

Proof. Proposition 7.1 would produce an E with shift at most zero, contrary to Theorem 6.2. □

These quantifiers are important. The corollary does not say that G_r has no LND at all. It neither makes an arbitrary derivation homogeneous in both gradings nor supplies the invariant required by the hypothesis.

8 Source checks and the global boundary

The source's degeneration uses

$$S = \mathbb{C}[p, x, y, z, u]/(xy - z(z+1) - p^3 u) \subset A, \quad R_0 = S/pS = \mathbb{C}[x, y, z, u]/(xy - z(z+1)).$$

This S is *not* the ambient kernel B of Theorem 3.2; the exponents $p^3 u$ and $p^2 t$ have different roles. With $s_0 = y - x(x - u^3)$, the center is $I = (x - s_0^2 - u^3, z - s_0 x) \subset R_0$. The source identifies

$$\operatorname{gr} A = R_0[\tau, I/\tau^2]$$

for its negative p-adic filtration. Its proof uses a saturated principal initial relation and replaces every bad representative by one of strictly smaller maximum weight. This termination statement, including negative weights, is retained; merely matching generic fraction fields would not identify the entire associated graded ring.

On the torsor (13), the source maps $x = ab, y = dc, z = db$ and principalizes I by $v = a^3 b - a^2 u^3 - d^2$. For example, with $h = x - u^3$ and $C_1 = c^2 - b^2 h$, it gives

$$z - s_0 x = b^2 v, \quad x - s_0^2 - u^3 = C_1 v,$$

and the explicit Bezout identity

$$\alpha C_1 + \beta b^2 = 1, \quad \alpha = a^2(1 - 2bd), \quad \beta = d^2(3 + 2bd) + \alpha h.$$

That last identity proves equality of ideals, not only containment. The pulled-back graded ring is therefore $R[\tau, V]/(\tau^2 V - v)$, the $r = 2$ case of our extraction lemma.

The main all-LND bridge still has separate content. A hypothetical polynomial structure on A is used to select a derivation with a positive-degree invariant. The source then forms its highest component, lifts the resulting additive action to a line bundle with a normalized cocycle, and checks compatibility with the other grading. This action need not preserve the original coefficient subring R_0 before lifting. An ambient slice with $\Delta H = p^3$, or the local statement $A_p \simeq \mathbb{C}[p^{\pm 1}]^{[3]}$, cannot replace these steps.

Corrections and retained valid details. No mathematical correction to the assigned source is asserted by this report's scoped reading. In particular we retain its exact cylinder divisibility, saturated initial relation, full principalization, action lifting rather than coefficient-preservation assumption, and use of the original rational function field for the final nonsquare. The source already contains the degree-two sharp example and the cubic obstruction; we do not relabel those as new. The larger odd-degree class, full ambient kernel, degree-m reduced inclusion, and higher-root extraction are the local increments actually proved here.

A literature-screening caution. The current survey [4] was inspected only selectively, and is not used as a proof premise. For example, its displayed derivation $x^2\partial_z - 2z\partial_y$ for the equation $x + x^n y + z^2 + t^3$ gives derivative $2z(x^2 - x^n)$, not zero for general n. This elementary check is sufficient reason not to import that display without its missing restriction. We make no full-survey audit claim. Recent metadata or a secondary status summary likewise cannot certify the assigned manuscript's new global assertion.

9 Advantages, limitations, and further work

Advantages. The full kernel has a fixed five-generator, one-relation presentation for every nonconstant ϕ, despite changes in the embedding and in the special-fiber degree. The saturation test reduces infinitely many possible p-denominators to one exact injection. The specialization theorem separates a finite cover on $X \neq 0$ from a missing boundary component; this is sharper diagnostic information than a transcendence-degree count. The orbit budget gives an explicit finite coefficient cutoff. The rigidity extension admits all lower auxiliary terms in its stated weight class, while the root lemma shows that square-killing survives every root exponent $r \geq 2$.

Limitations. The kernel result concerns one chosen ambient LND, not all actions on a hypersurface quotient. The rigidity theorem requires exact homogeneity, odd degree at least three, and an exact equation $E^2 v = 0$. A small numerical residual cannot be substituted. The root lemma requires an actual domain, a positive homogeneous invariant, and compatibility with two gradings. Neither it nor the odd-degree extension verifies the existence of geometric filtrations or line-bundle lifts for a new family. The polynomial *abc* argument, Gauss's lemma, local slices, and saturation are classical tools; their names or finite numerical reproductions are not novelty claims.

Feasible next questions. One can classify the branch and missing-divisor behavior of the explicit map $\operatorname{Spec} \mathbb{C}[s, u, L] \to \operatorname{Spec} B_0$ for chosen ϕ, keeping global surjectivity separate from finite behavior on a chart. A second problem is to determine which lower auxiliary perturbations admit the complete source degeneration and torsor construction, rather than only the local obstruction. Even degree requires a different invariant: the explicit square-zero leading-form shear shows why the present valuation proof cannot simply be reused. Finally, exact elimination certificates could be produced for concrete finite coordinate banks, with separately proved degree bounds; sampled ranks alone would not supply completeness.

10 Evidence and contribution assessment

The owned read ledger records each actual numbered display interval, its complement, and pre/post SHA-256. The complete public scope is ten TeX files plus README, 2,299 lines. The private scope is 150 selected lines: 48 in the original finite-moment manuscript and 102 in the specific

slice/saturation comparison. The remaining 1,570 lines of those two files, all other confidential corpus material, and uninspected external proofs are not credited as read. Public web requests used only public titles or URLs. No confidential original was uploaded.

The independently executable owned test suite checks exact polynomial identities, finite orbit orders and synthesis, fiber-coordinate formulas, the sharp and even-degree examples, and bounded graded-piece and degree-inequality instances. It passed 30,985 exact assertions in 23,497 bounded cases; its JSON gives the separate units and domains. Such tests supplement the proofs; they do not establish an all-LND classification, an unbounded saturation assertion by enumeration, or a global cancellation theorem. Parent reviewer checks are kept separate and are not included in author counts.

Applicability remains L3. The contribution is assessed independently as C3: the concrete global kernel and quantified failure of specialization are a substantive source-specific algebraic calculation, and the odd-degree/lower-term and higher-root statements are modular extensions with complete local proofs. This is not C4 or C5: no new general geometric lifting theorem, full independent global cancellation proof, or broad all-LND classification is supplied. The proof boundaries and classical attributions are part of the assessment, not qualifications to be removed after successful compilation.

References

[1] *An explicit failure of complex affine-space cancellation*, assigned public manuscript, September 23, 2026. Cached source snapshot in the project, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; ten TeX files and README. Parent captured-target authentication is recorded separately in `_audit/batch44-source-identity.json`; it is snapshot provenance, not proof certification. The owned ledger records independent local hashes.

[2] S. Gaifullin and M. Petrov, *Non-cancellative varieties, maximal tori, and the Makar-Limanov invariant*, July 2026 preprint, arXiv:2607.13593. Primary introduction and selected local-lifting discussion inspected; not a full proof audit.

[3] A. K. Dutta and A. Lahiri, *On Residual and Stable Coordinates*, arXiv:1908.03549. Theorem 3.4 and proof, with its hypotheses, inspected in the primary text.

[4] C. F. Venegas and H. J. Venegas, *Cancellation problem via locally nilpotent derivations*, arXiv:2512.15590v2, February 17, 2026. Selected statements screened; not relied on as a theorem input. See Section 8.

[5] M. Subbarao, confidential original finite-moment manuscript, `TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, lines 155–180 and 455–476 freshly inspected. Finite analysis and positive weighted Gram construction only; no whole-corpus reading claim.

[6] M. Subbarao, confidential local slice and saturation comparison, `res-math/TR-item-061-astra.tex`, lines 290–364 and 398–424 freshly inspected. This is a specific comparison, not a replacement for the original in the preceding entry.